MILITARY SPECIFICATION

COMPASS SYSTEM: MAGNETIC SLAVED,
DIRECTIONAL GYRO STABILIZED, 4°/HR, TYPE MA-1

This amendment forms a part of Military Specification MIL-C-17858C(AS), dated 1 June 1966, and has been approved by the Naval Air Systems Command, Department of the Navy.

Page 1, at bottom,

Add the following new statement:

COMMENTS OR RECOMMENDATIONS WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT MAY BE SENT TO THE PREPARING ACTIVITY WHEN A SELF-ADDRESSED STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL (DD FORM 1426) APPEARS AT THE END OF THIS DOCUMENT, OR BY LETTER ADDRESSED TO THE COGNIZANT ACTIVITY.

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* 3.5.2 Controller - Delete first sentence and substitute new first sentence as follows:

"The controller shall be in accordance with MS90316, MS90319 or MS14173 as applicable"

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Add the following new paragraphs:

3.5.9 Lubrication -

3.5.9.1 Rotor Bearing Lubrication - The rotor bearings shall be lubricated with grease conforming to MIL-G-81322 or MIL-G-81937 with particle size not exceeding 50 microns.

3.5.9.2 Bearing Lubrication - The lubricating oil used in gimbal bearings shall be MIL-L-81846.

3.5.9.3 Barrier Film Bearing Coating - All bearings shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with MIL-B-81744. All bearings, including chamfered races or corners, and including interfacing parts where applicable, shall be barrier coated.

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3.5.9.4 Ball Bearings - Ball bearings shall be ABEC-5 grade or better. Teflon separators shall be used on gimbal bearings where the bearing design permits. Spin rotor bearings shall have porous separators impregnated with lubricant MIL-L-81846 and shall be ABEC-7 or better.

3.5.9.5 Ball Bearing Pre-Loading - Means shall be provided for preloading and alignment without causing eccentric loading unless otherwise authorized.

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4.4.3: Add the following to end of paragraph:

Classification of a failure shall be in accordance with MIL-STD-781 and AR-34.

Add the following new paragraph:

4.4.3.1.1 Reliability Assurance Production Acceptance Phase - Reliability Assurance Production Acceptance Testing shall be conducted through a production burn-in. Each equipment producted but not submitted to the Qualification Test shall be operated for 150 operating hours burn-in, and the test data (operating hours and failures) shall be totaled and compared to the Accept/Reject Criteria of Test Plan II of MIL-STD-781. (Extend the lines as necessary to accomodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. As the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. Delivery of equipment shall not be held up to await a decision under the test plan. In event an Accept Decision is reached, the test duration of each equipment shall be reduced to 100 operating hours. At any time that the current total of test hours and test failures plotted on Test Plan IV curves (extended) show a Reject Situation, the Procuring Activity shall be notified. The Procuring Activity reserves the right to stop the acceptance of equipment at any time that a Reject Situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a rejection decision. For the Production Acceptance Phase, Test Level E shall be used, except that the temperature extremes shall be +55C and -44C, and vibration shall be 0.5G. Details of the burn-in tests shall be included in the test procedures.

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* 6.8 Items of Equipment - Add the following:

Component	Military Standard System A
"Controller MA-1 Compass	MS14173"

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6.10: Add the following under military specifications:

MIL-G-81322	Grease, Aircraft, General Purpose Wide Temperature Range
MIL-B-81744	Barrier Coating Solution and Solvent
MIL-L-81846	Lubricating Oil, Instrument, Ball Bearing, High Flash Point
MIL-G-81937	Grease, Instrument, Ultra Clean

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Add the following under military standards:

MIL-STD-1334 Process for Barrier Coating of Anti-Friction Bearings

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* 6.10: Add the following:

Military (Continued)

MS14173	Controller, MA-1 Compass (A-6 aircraft only)

Naval Air Systems Command

AR-34	Failure Classification for Reliability Testing, General Requirements for

Custodian:
Navy -Navy - AS

Preparing Activity:
Navy - Navy - AS
Project No. 6605-N280

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